Geocaverns: A Machine Learning Evaluation Of Cave Tourism And Biodiversity

Given Name Surname
dept. name of organization
(of Affiliation)
name of organization
(of Affiliation)
City, Country
email address or ORCID

ABSTRACT

The assessment of biodiversity and protection of natural ecotourism sites depend on our knowledge of underground environments but cave systems face difficulties because of their limited data access and their intricate earth surface patterns. The research presents GeoCaverns as a complete geospatial system which uses machine learning to study and forecast ecological developments found in cave environments. The framework uses clustering and regression and anomaly detection and recurrent neural networks to create models which represent biodiversity and accessibility and depth attributes from a synthetic global cave dataset that maintains structural integrity. The system achieves spatial continuity through its 3D surface modeling method which uses depth interpolation while users can view geospatial data through the GIS-based interface which supports interactive exploration. The research shows that machine-learning models effectively support the assessment of subterranean ecosystems which require data-driven methods for conservation and management purposes.

I. INTRODUCTION

The environmental system of cave ecosystems exists as a distinctive natural system which develops through three main processes: geological, hydrological, and ecological processes. The system functions through its intricate internal framework which allows limited sunlight to enter while creating isolated climates that enable the development of specialized biological systems which face high risks from both natural and human-related changes. The field of subterranean research has not reached scientific saturation. According to Gunn [1] and Culver and Pipan [2] researchers encounter major obstacles when studying these environments because they require special methods to access remote areas which make it hard to gather data and perform extended ecological studies. The research results lead to ongoing knowledge shortfalls about how species distribute themselves in caves and how human activities impact cave environments and how human activities affect cave ecosystems.

The geospatial analytics and machine learning field has obtained new developments that can help researchers solve pressing problems. Spatial data mining techniques have demonstrated effectiveness in identifying patterns across geographically distributed natural systems which enables better environmental assessment [5]. Machine learning based environmental modeling shows strong potential to forecast ecological and geomorphological conditions in areas with insufficient data [3]. The technology has various applications which include species-distribution modeling and structural parameter estimation. Researchers have not started applying complete computational techniques to cave study despite technological progress. The existing research about karst and subterranean environments requires more comprehensive datasets because researchers study particular areas through one research method which restricts their ability to draw common ecological conclusions about these environments[10].

Machine learning frameworks such as clustering algorithms, anomaly detection methods, and regression-based predictive models have shown value in identifying structural patterns and ecological anomalies within multidimensional environmental datasets [6, 7]. The latest advancements in deep learning, especially through recurrent neural networks (RNNs), now enable precise predictions of ecological time series data throughout various environmental conditions including noisy and nonlinear situations [4, 9]. The current situation lacks any consolidated platform that combines these analytic methods into a complete system for evaluating, displaying, and predicting cave ecosystem dynamics.

The present study introduces GeoCaverns as a solution to this problem by developing an integrated geospatial machine learning framework which evaluates cave ecosystems through their structural and ecological features. The system includes KMeans and Agglomerative Clustering and Birch as its clustering algorithms, while it uses Gradient Boosting and Random Forest and Elastic Net as its supervised regression models and Isolation Forest and LOF as its anomaly detection algorithms and SimpleRNN for its forecasting system. The depth surfaces and spatial cave distribution visualization use geostatistical interpolation techniques [11] and GIS-based mapping tools [10] for their construction. GeoCaverns establishes an all-inclusive research platform which allows users to forecast species resources and assess tourism routes and discover structural irregularities and research spatial depth changes.

This research adds to the expanding field of data-driven geoscience by showing how scientists can use multimodal machine learning together with geospatial modelling to study subterranean ecosystems. The research framework provides a

scalable and interactive system which enables researchers to conduct cave studies while gaining better ecological understanding and developing conservation strategies for sensitive underground environments. The proposed system uses spatial analytics together with predictive modeling and three-dimensional reconstruction to create a solution which connects environmental modeling research with practical decision-support systems. The research establishes a basis for future studies which will integrate geoscience and ecology with artificial intelligence to improve sustainable management of subterranean resources.

II. METHODOLOGY USED

The methodology of this study provides information about the used dataset and the developed analytical framework and the employed hardware and evaluation metrics. Subsections II(A), II(B), and II(C) present the dataset description, system framework, and performance evaluation criteria, respectively. This section describes the resources and tools and procedures which is used to study cave ecosystems and forecast biodiversity and tourism patterns.

A. DATASET

The proposed machine learning models were evaluated through training and testing which used the GeoCaverns Cave Ecosystem Dataset. The dataset consists of synthetic cave records which represent various geographical areas and different ecosystem environments. The record includes numerical and categorical data points which show the following information: latitude, longitude, depth (m), cave length (km), cave type, access difficulty, species found, and tourist accessibility. The cave ecosystem attributes show the geological structure and biological diversity and tourism potential of the cave system.

The dataset creates synthetic data which models real cave ecosystem features because actual cave data has limited availability and accessibility and completeness. The synthetic data permits controlled tests which produce equal geological and ecological results while enabling to verify their findings through testing which follows standard procedures.

To enhance model performance, derived features were computed from the raw attributes. A composite **Ecosystem Index (EI)** was formulated to represent the combined influence of geological and biological factors and is defined as:

$$EI = \frac{D + L + S}{3}$$

where D denotes cave depth (m), L represents cave length (km), and Sindicates the number of species found.

Additionally, a **Tourism Accessibility Score (TAS)** was derived to reflect the suitability of caves for tourism activities and is expressed as:

$$TAS = \alpha A + \beta(1 - ND)$$

where Ais the binary tourist accessibility indicator, ND is the normalized depth value, and α and β are weighting constants controlling the contribution of accessibility and depth.

Numerical encoding methods is employed to transform cave type and access difficulty which both represented categorical variables into numerical data. The normalization methods is applied to create standardized continuous attributes which they used in multiple modeling activities that included clustering and regression and anomaly detection and biodiversity forecasting.

Table 1 shows a statistical summary of the GeoCaverns dataset which presents average values and standard deviation values for all features to demonstrate how spatial distribution and geological diversity and biodiversity distribution between caves. Table 2 shows the distribution of categorical attributes through frequency and percentage analysis. Table 2(A) describes the distribution of cave types which shows an equal representation of geological formations through the complete display of Ice Caves and Limestone and Lava Tubes and Solutional and Sea Caves and Karstic caves which effectively classifies ecosystems. The access difficulty distribution shows an equal distribution of Easy Moderate Hard and Extreme categories which match the actual limits of exploration and accessibility. Table 2(C) provides a summary of tourist accessibility which indicates that tourist-accessible caves and non-accessible caves exist in equal proportions which creates equal learning opportunities for tourism prediction and classification tasks.

TABLE I: Numerical Features of The GeoCaverns Dataset

S. No	Characteristic	Description	Mean ± Std
1	Latitude (F1)	Geographical latitude of cave locations	**0.07 ± 34.84**
2	Longitude (F2)	Geographical longitude of cave locations	**−2.68 ± 103.40**
3	Depth (m) (F3)	Vertical depth of caves	**1013.38 ± 574.45**
4	Length (km) (F4)	Total mapped cave length	**150.97 ± 86.09**
5	Species Found (F5)	Number of species identified	**50.26 ± 29.30**
6	Cave Type (F6)	Encoded geological cave type	**2.46 ± 1.72**
7	Access Difficulty (F7)	Encoded cave access difficulty	**1.51 ± 1.12**
8	Tourist Accessible (F8)	Tourism accessibility indicator (0/1)	**0.50 ± 0.50**

Note: Mean ± Std is reported only for numerical attributes.

TABLE II : Cave Type Distribution

Cave Type	Count	Percentage (%)
Ice Cave	800	17.78
Limestone	756	16.80
Lava Tube	754	16.76
Solutional	734	16.31
Sea Cave	730	16.22
Karstic	726	16.13

TABLE III: Access Difficulty Distribution

Access Difficulty	Count	Percentage (%)
Extreme	1160	25.78
Moderate	1154	25.64
Easy	1108	24.62
Hard	1078	23.96

A. Tourist Accessibility Distribution

Tourist Accessible	Count	Percentage (%)
No	2263	50.29
Yes	2237	49.71

Categorical attributes are represented using frequency and percentage distributions. Numerical encoding was applied only for machine learning model implementation.

III. PROPOSED FRAMEWORK

The GeoCaverns system framework starts with data pre-processing as its first stage and moves to feature extraction before proceeding to modeling and evaluation. Pre-processing functions as an essential step because the input data quality and structure determine how well machine learning models will perform in analyzing cave ecosystem data.

A. PRE-PROCESSING

The proposed framework uses a complete pre-processing system to process raw cave ecosystem data which produces dependable results that scientists can use for their modeling work. The data processing stage helps to increase the accuracy of classification and clustering and prediction results because machine learning models depend on data quality. The pre-processing stage includes outlier detection and removal (P), handling missing values (Q), feature scaling and normalization (R), and feature selection (F), as described below.

1) OUTLIER DETECTION AND REMOVAL

The proposed framework uses a complete pre-processing system to process raw cave ecosystem data which produces dependable results that scientists can use for their modeling work. The data processing stage helps to increase the accuracy of classification and clustering and prediction results because machine learning models depend on data quality. The pre-processing stage includes outlier detection and removal (P), handling missing values (Q), feature scaling and normalization (R), and feature selection (F), as described below.

$$P(x) = x, \qquad Q1 - 1.5 \times IQR \leq x \leq Q3 + 1.5 \times IQR$$

where x represents an individual feature value in the n-dimensional feature space, while Q1, Q3, and IQR define the first quarter, third quarter, and interquartile range of data distribution. The method protects all geological and ecological patterns in the dataset from distortion through extreme value handling.

2) MISSING VALUE HANDLING

After outlier removal, the dataset was examined for missing or invalid values, which can mislead machine learning models and degrade predictive performance. Instead of discarding incomplete records, missing values in continuous attributes were replaced using **mean value imputation**, as defined in:

where x represents an attribute value in the feature vector. Mean imputation preserves the overall data distribution and prevents the introduction of artificial outliers, making it suitable for continuous cave-related measurements such as depth and length.

3) FEATURE SCALING AND STANDARDIZATION

Distance-based machine learning models and gradient-optimization machine learning models require feature scaling as an essential process. The study used Z-score normalization to standardize continuous numerical attributes according to their distribution patterns, as expressed in:

$$R(x) = \frac{x - \bar{x}}{\sigma}$$

where $\bar{x}$ and σ represent the mean and standard deviation of the feature, respectively. The process of standardization establishes a zero mean value and a unit variance value while it decreases data skewness and helps models to train more effectively. The research shows that some tree-based models demonstrate low responsiveness to changes in feature scaling.

4) FEATURE SELECTION

As feature space dimensionality rises, machine learning models lose their capacity to generalize due to the curse of dimensionality. High-dimensional feature spaces and small sample sizes combine to produce circumstances that cause overfitting because there is not enough data to accurately depict real patterns. To lessen modeling effects, the modeling process needed a selection of geological, biological, and tourism-related features that demonstrated real value. Based on their statistical significance and the expertise of subject-matter experts, pertinent features were chosen.

In order to support all subsequent stages, such as feature extraction, clustering, regression modeling, anomaly detection, and biodiversity forecasting, the GeoCaverns framework employs a pre-processing step that produces a clear and consistent data representation that contains important information.

B. EXPLORATORY DATA ANALYSIS (EDA)

Before creating the model, the GeoCaverns dataset's statistical, spatial, and ecological properties were investigated using exploratory data analysis, or EDA. Finding data distributions, variable relationships, and possible anomalies that could affect machine learning performance is made possible by this step. Informed choices about feature selection, scaling, and the analytical methods used in later framework stages are further supported by visual exploration.

The Species Found attribute was analysed using a histogram to look at biodiversity variability across cave ecosystems. The observed distribution which shows a broad range of species counts demonstrates that ecological diversity among caves. The existence of both high-diversity and low-diversity caves shows that different environmental conditions exist which makes regression-based prediction models more effective than simple threshold-based classification methods. The distributional analysis is conducted for both depth and cave length attributes which revealed substantial differences that justified their choice to use normalization and standardization methods during data pre-processing.

The study used scatter plots to analyse spatial relationships between cave locations and geological structures and their associated biodiversity. The scatter plots which used different colors to represent species counts showed that biodiversity distributed itself across geographic regions instead of spreading out into random patterns. The three-dimensional visualizations show how depth changes throughout the cave system which proves that vertical cave structure determines which subterranean ecosystems develop within the space. The spatial observations from the study created a research need to apply clustering and spatial learning methods for discovering natural cave groupings which use both geological and ecological data.

Principal Component Analysis (PCA) is used to reduce dimensionality of standardized numerical features which enabled visualization of major variance patterns present in the dataset. The partial separation between cave types which projection onto the first two principal components showed establishes that geological categories keep their unique structural patterns in the reduced feature space. The study demonstrates that geological features play a vital role in ecosystem description which justifies their use in both supervised prediction methods and unsupervised clustering applications.

The GIS-based mapping techniques is used to create geospatial visualizations which improved spatial understanding and enabled users to explore the data intuitively. The latitude and longitude coordinates were used to create interactive geographic maps which displayed cave locations through markers that showed individual caves and their associated biodiversity data. The maps provide visual detection of spatial clusters and regional biodiversity patterns and potential spatial anomalies while functioning as an interactive decision-support system within the GeoCaverns framework for conservation planning and tourism management.

C. MACHINE LEARNING PIPELINES

The GeoCaverns framework uses multiple machine learning pipelines to study cave ecosystem characteristics through geological biological and spatial data. The pipelines enable the machine learning system to perform learning tasks which include ecosystem classification and biodiversity prediction and anomaly identification and temporal forecasting. The framework uses unsupervised learning and supervised learning and sequential learning methods to create an analytical method that can detect complex nonlinear connections and concealed patterns in underground data.

1. CLUSTERING MODELS

The unsupervised clustering models is used to discover natural group patterns in cave ecosystems which were based on their geological and spatial and ecological characteristics. The numerical features are standardized before starting the clustering process to achieve consistent scaling which would stop high-value attributes from dominating the results. The multiple clustering algorithms is tested which included K-Means and Agglomerative Hierarchical Clustering and BIRCH because each method provided unique advantages for managing different types of data which included worldwide data distribution and hierarchical structures and extensive datasets. The Fuzzy C-Means clustering is used to create a model which showed how different ecosystems shared their borders while Self-Organizing Maps (SOM) maintained topological structure to show hidden spatial patterns. These clustering methods are used to create ecosystem-level interpretations which identified caves with matching structural patterns and biodiversity patterns to establish a foundation for ecological comparison studies and visual representation.

2. REGRESSION MODELS

Supervised regression models were used to forecast continuous cave ecosystem characteristics which included biodiversity levels and geological depth. Using spatial and structural features as input, five distinct regression models—including Linear Regression, Random Forest Regression,

Gradient Boosting Regression, K-Nearest Neighbors, and LightGBM—are trained. Tree-based ensemble models successfully captured the nonlinear interactions which existed between cave depth and length and geographic location. The error-based metrics is used to evaluate model predictions because these metrics provide information about both accuracy and generalization performance. The regression pipelines enable the estimation of ecological indicators through quantitative methods while they use scenario-based analysis to support conservation and tourism planning.

3. MULTI-OUTPUT PREDICTION

Interconnected cave ecosystem attributes are modelled using multi-output prediction techniques. Using shared feature representations, the Multi-Output Decision Tree Regression and Multi-Task Elastic Net Regression are used to predict several targets, such as species count and cave depth. In order to improve predictive performance beyond what single-output models can accomplish, the method takes into account multiple output variables. Holistic ecosystem modelling is made possible by multi-output learning, which combines biological and geological processes into a single predictive framework.

4. ANOMALY DETECTION

To find rare, unusual, or structurally unique cave records that might indicate extreme geological formations, special hotspots for biodiversity, or possible data inconsistencies, anomaly detection techniques were used. Without using labelled examples, anomalies in multidimensional feature space were found using the Local Outlier Factor (LOF) and Isolation Forest algorithms. These models use local density deviation and data isolation, respectively, to determine anomaly scores. Spatial visualization was used to further analyse the anomalies found in order to determine their ecological and geological significance. This element facilitates the early detection of caves that need extra research or special conservation care.

5. TEMPORAL FORECASTING

The GeoCaverns pipeline used sequential learning models to predict cave biodiversity changes and track biodiversity changes through time. Simple recurrent neural networks (RNNs) learned to process species count sequences that contained nonlinear time dependencies. Cellular Automata models simulated spatial propagation patterns and showed how localized interaction effects operated, while Markov Chain models showed how biodiversity states transitioned between different discrete states. The two temporal modelling methods work together to provide trend analysis and short-term forecasting while showing how ecosystems will respond to human activities and environmental changes.

IV. EVALUATION METRICS

The evaluation of the GeoCaverns framework performance used different metrics which focused on specific tasks to measure the framework's abilities in regression accuracy, clustering quality, anomaly detection, and model stability tests. The framework required multiple evaluation methods because it combined supervised learning, unsupervised learning, and temporal learning methods which needed different assessment methods for their analytical work.

Standard error-based metrics is used to evaluate regression-based prediction tasks which included biodiversity estimation and geological depth prediction. The Root Mean Square Error (RMSE) metric assessed prediction error magnitude while it imposed higher penalties on larger errors, which made it effective for detecting significant ecological prediction errors. The Mean Absolute Error (MAE) showed the average prediction error in a simple way, but the R^2 score explained how much of the actual data variance the model could predict. The metrics enabled researchers to evaluate the predictive accuracy and generalization ability of different regression models through their combined use.

The unsupervised clustering results were evaluated through internal validation metrics because actual cluster labels needed for evaluation were not available. The Silhouette Score evaluated the cluster performance by measuring their internal similarities and external distances to determine how well they maintained their internal structure while separating from each other. The Calinski–Harabasz (CH) Index assessed the relationship between between-cluster variance and within-cluster variance which helped to determine how tightly clusters formed and how well they separated from each other. The Davies–Bouldin (DB) Index measured average cluster similarity between clusters where lower values demonstrated better cluster differentiation. The combination of these metrics establishes strong proof about how well clustering performs in the cave ecosystem environment.

The reconstruction-based error analysis was used to execute its two main tasks which were anomaly detection and representation learning. The autoencoder reconstruction error measures mean squared difference which compares original feature vectors to reconstructed feature vectors and shows how well normal cave patterns had been learned. High reconstruction error values enable identification of potential anomalies and structural differences in cave formations. This evaluation strategy works together with statistical anomaly detection methods because it detects nonlinear data space patterns.

The model comparison tables and visualization plots are created to establish comprehensive evaluation methods that enable direct assessment of various educational methods. The RMSE, MAE, R^2, and clustering scores are used to compare different models, which resulted in finding the best models for each task. The visual tools, which include prediction-versus-observation plots and error distribution charts and cluster visualization graphs, helped assess quality and make things understandable. The multi-metric evaluation method establishes that chosen models must demonstrate equal performance across multiple areas while maintaining strength

and environmental applicability within the GeoCaverns framework.

V. SPATIAL MODELING

The GeoCaverns framework was further extended to incorporate spatial modeling to address the three-dimensional nature of cave systems and the spatial continuity of geological depth over geographical locations. The three-dimensional nature of cave systems and the spatial nature of cave systems imply that two-dimensional analysis is insufficient to capture the complexity of cave systems. The proposed framework integrates geostatistical interpolation techniques and three-dimensional surface extraction algorithms to enhance spatial analysis and visualization of cave systems.

Depth interpolation was performed to estimate continuous depth surfaces from a set of discrete measurements of cave depth. The grid-based spatial interpolation technique was applied to estimate depth values on a spatial grid to produce a smooth three-dimensional surface map of depth. Linear interpolation was applied to ensure spatial continuity, and nearest-neighbor interpolation was applied to assign values to regions of sparse data. The resulting 3D depth maps enable the interpretation of subsurface variability and visualization of regional depth gradients, which are difficult to determine from point data.

The Marching Cubes algorithm serves as the foundation of the three-dimensional surface reconstruction method which improves the assessment of the cave system's three-dimensional architectural dimensions. The Marching Cubes algorithm creates polygon mesh structures through the extraction of iso-surface information which it derives from depth field-based volumetric data that has been interpolated. The resulting mesh geometries enable the analysis of periodic geometric pattern designs and depth variations of the underground caves which acts as a tool for visualizing the underground rock formations. The Marching Cubes algorithm enables to analyze geometric patterns of caves using different depth analysis methods that enable comparisons of different cave systems.

The GeoCaverns system uses depth interpolation and three-dimensional surface reconstruction methods to develop a spatial modeling system that enables accurate and interpretable models of the geometry of the caves. Spatial models enable machine learning predictions to include geographic and structural knowledge which enhances ecological interpretation and decision-making processes for cave conservation and sustainable tourism management.

VI. INTERACTIVE DEPLOYMENT

The GeoCaverns framework was developed as an interactive web-based decision-support system which operates on the Streamlit platform to achieve its goal of practical usability and real-world applicability. The interactive system enables researchers and conservation planners and tourism managers to access cave ecosystem insights through its machine learning analytics which were designed to be user-friendly. The deployment environment enables users to create dynamic data visualizations while they conduct real-time model inference and navigate the framework's analytical components.

The Streamlit-based user interface through its sidebar-driven workflow enables users to access the different analytical modules of GeoCaverns in an organized manner. Users can view raw and processed datasets, explore exploratory data analysis outputs, and interact with clustering, regression, anomaly detection, and spatial modeling results. Input forms allow users to specify attributes of a new or hypothetical cave, including geographic coordinates, depth, length, cave type, and access difficulty. The system automatically processes user inputs and converts them into the feature representations which were utilized during the model training process.

Machine learning models that use trained algorithms run in real time after users submit their input to produce predictive results. The system generates ecosystem cluster assignments which include estimated biodiversity levels and indicators of tourism accessibility and short-term biodiversity forecasts that apply to relevant situations. The system enables model inference to be executed within its deployment environment which provides instant results and allows users to test different cave designs. The system enables real-time predictions which improve scenario analysis results and enable data-driven decision-making for conservation and tourism planning activities.

The interactive deployment system depends on visualization as its main component. The dashboard uses two-dimensional and three-dimensional plots together with geospatial maps and statistical charts to display model outputs in a way that users can easily understand. Geographic visualizations show cave locations together with their associated spatial patterns while three-dimensional surfaces and scatter plots show changes in depth and structural connections. The interactive components create visual elements which transform difficult analytical results into practical insights while the GeoCaverns framework functions as a user-friendly platform which can grow and develop for research into underground ecosystem systems.

VII. RESULTS AND DISCUSSION

The experimental results of the GeoCaverns framework testing process are presented in this section. The results analysis evaluates two aspects, which are structural ecosystem grouping and clustering validation, and three testing methods, which include regression and forecasting and three-dimensional spatial modelling effectiveness.

A. STRUCTURAL ECOSYSTEM REPRESENTATION

For analyzing the inherent structure of cave ecosystems, Principal Component Analysis (PCA) was coupled with KMeans clustering. The projection obtained is depicted in

Fig. 1. By projecting the high-dimensional geological and ecological feature space onto the two principal components, the variance patterns are made visible. As seen in Fig. 1, the projected clusters are partially geometrically separated with smooth transitions between the boundaries. This suggests that the cave ecosystems have overlapping structural properties rather than being strictly separated. The cluster structure obtained is consistent with the fact that ecosystem similarity is determined by multidimensional interactions between latitude, longitude, depth, and biodiversity variables. The PCA-KMeans projection thus provides a compact structural description of subterranean ecosystem clustering.

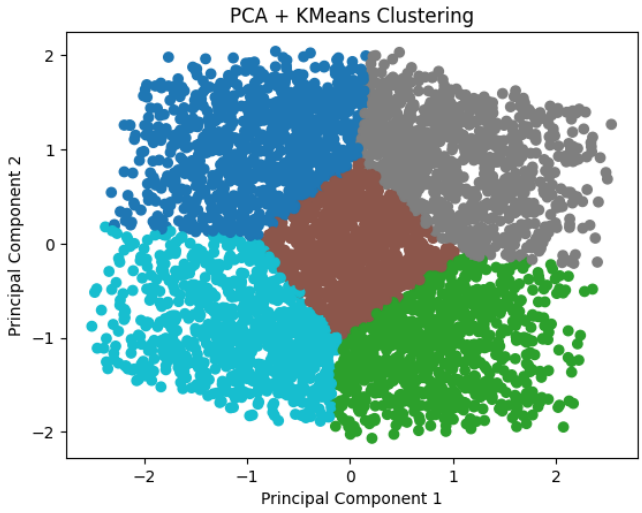


Fig. 1 represents the reduced-dimensional clustering structure of cave ecosystems derived from standardized geological and ecological features.

B. CLUSTERING VALIDATION AND UNSUPERVISED LEARNING PERFORMANCE

The performance of the clustering algorithm was evaluated using Agglomerative Clustering, Fuzzy C-Means, and Birch algorithms. The internal validation metrics Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index were used to evaluate the compactness and separation of the clusters.

The maximum Silhouette Score (0.087) and Calinski-Harabasz Index (475.760) were achieved by the Birch clustering algorithm, which represents relatively better separation between the clusters. The minimum Davies-Bouldin Index (1.888) was achieved by the Fuzzy C-Means clustering algorithm, which represents relatively better intra-cluster similarity. Although the Silhouette scores are moderate, these are in line with the gradual ecological transition patterns found in natural subterranean ecosystems. The clustering analysis clearly reveals the presence of strong multidimensional ecosystem stratification, which is not based on geographic stratification.

C. REGRESSION AND MULTI-TARGET PREDICTION RESULTS

Regression performance on various models is presented in Fig. 2. The RMSE comparison allows for the direct assessment of biodiversity and depth prediction accuracy on various models.

In biodiversity prediction, Multi-Task Elastic Net has the lowest RMSE of 30.119, marginally better than LightGBM with an RMSE of 30.972, whereas Decision Trees have a higher error of 42.399. In depth prediction, Elastic Net substantially decreased the error to 573.411, which is significantly better than Decision Trees with an error of 828.769.

Elastic Net, as evident from Fig. 2, maintains a consistently lower prediction error on both targets, thereby ascertaining that regularized linear modelling is effective in identifying the structural patterns in the dataset.

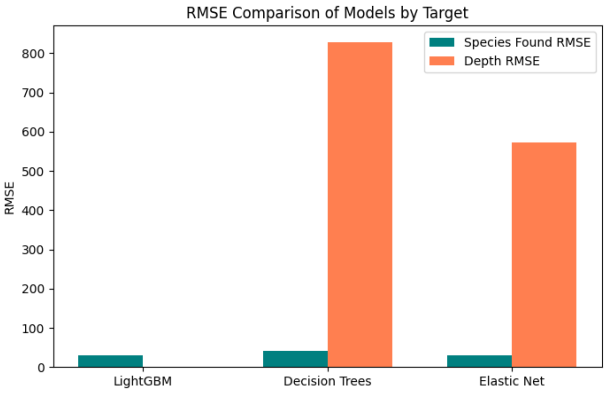


Fig. 2 represents the RMSE-based performance comparison of regression models for biodiversity and depth prediction tasks.

D. TEMPORAL FORECASTING EVALUATION

The experiments on temporal biodiversity forecasting also probed the capability of dynamic modeling. The Simple Recurrent Neural Network had the lowest RMSE of 31.175, performing better than Cellular Automata with an RMSE of 39.874 and the Markov Chain model with an RMSE of 44.263. The findings of the forecasting experiment highlight the significance of neural sequence learning in modeling the dynamic biodiversity indicators in cave ecosystems.

E. REPRESENTATION LEARNING AND THREE-DIMENSIONAL SPATIAL MODELING

The experimental results demonstrated that nonlinear representation learning outperformed all other methods because it maintained better integrity of structural data compared to different methods.

The Autoencoder achieved better reconstruction performance through its mean squared error measurement of 0.2655,

which exceeded the PCA method that showed a 0.3213 reconstruction error because it could develop a nonlinear model. The Marching Cubes algorithm produced the volumetric reconstruction, which serves as the basic component that builds the subterranean structure model. The generated output creates a three-dimensional surface mesh that appears in Fig. 3. The mesh representation introduces a new method to show how depth changes throughout the structure, which differs from the traditional 3D scatter plot method. The algorithm created a polygonal surface that features 1434 extracted vertices and 2624 extracted faces, which combined to show the volumetric and spatial structure.

The reconstructed mesh shows how depth changes throughout the cave formation in Fig. 3, while maintaining the structural integrity of the cave system. The GeoCaverns system uses this volumetric modeling method to create spatial components that set it apart from traditional two-dimensional geospatial analysis methods.

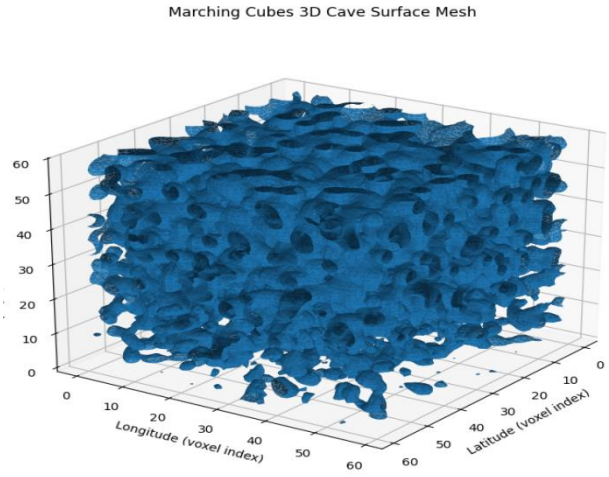


Fig. 3 represents the reconstructed three-dimensional cave surface generated using the Marching Cubes algorithm.

The experimental results through their testing demonstrate that GeoCaverns successfully merges three separate analytical methods into one complete system through its combination of unsupervised clustering with multi-target regression and temporal forecasting and volumetric reconstruction functions. The PCA-based structural grouping method offers interpretable ecosystem segmentation while Elastic Net provides strong predictive performance and recurrent neural networks enhance their ability to model time-dependent data. The three-dimensional surface reconstruction process allows researchers to better understand both the spatial characteristics and the structural elements of the object being studied.

The research outcomes demonstrate that the combination of geospatial analytics with multimodal machine learning methods effectively assesses cave ecosystems while supporting sustainable management planning.

VIII. FUTURE WORK

Although the GeoCaverns framework shows the potential of geospatial analytics and multimodal machine learning in the assessment of cave ecosystems, there are still some areas of extension that can be explored. Firstly, the current study is based on a structurally consistent synthetic dataset. The next step will be to validate the framework using real-world datasets of cave surveys, including speleological, environmental sensor, and biodiversity data. This will allow a more precise ecological validation of the framework.

Secondly, more complex models of deep learning, such as Long Short-Term Memory (LSTM) networks or Graph Neural Networks (GNNs), could be used for the better modeling of complex temporal dependencies and spatial relationships between the interconnected cave systems.

The spatial modeling component can be expanded through the integration of high-resolution LiDAR scans and ground-penetrating radar data which enables the creation of more accurate volumetric cave models. The combination of geostatistical interpolation with physics-informed geological simulations will enhance the accuracy of three-dimensional structural modeling.

The semi-supervised learning techniques together with uncertainty quantification methods will improve the reliability of the anomaly detection framework which identifies ecologically sensitive areas and structurally uncommon formations. The interactive deployment will develop into a cloud-based decision-support platform which provides real-time data access to support ongoing monitoring and flexible tourism management operations. GeoCaverns will evolve from its current state as a predictive modeling system into a full-fledged intelligent management solution for underground environmental protection.

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Relevant to depth interpolation and grid-based spatial modelling.